Frequency Attack on Monoalphabetic Substitution Cipher

40. Write a C program that can perform a letter frequency attack on any monoalphabetic substitution cipher without human intervention. Your software should produce possible plaintexts in rough order of likelihood. It would be good if your user interface allowed the user to specify "give me the top 10 possible plaintexts."

Aim

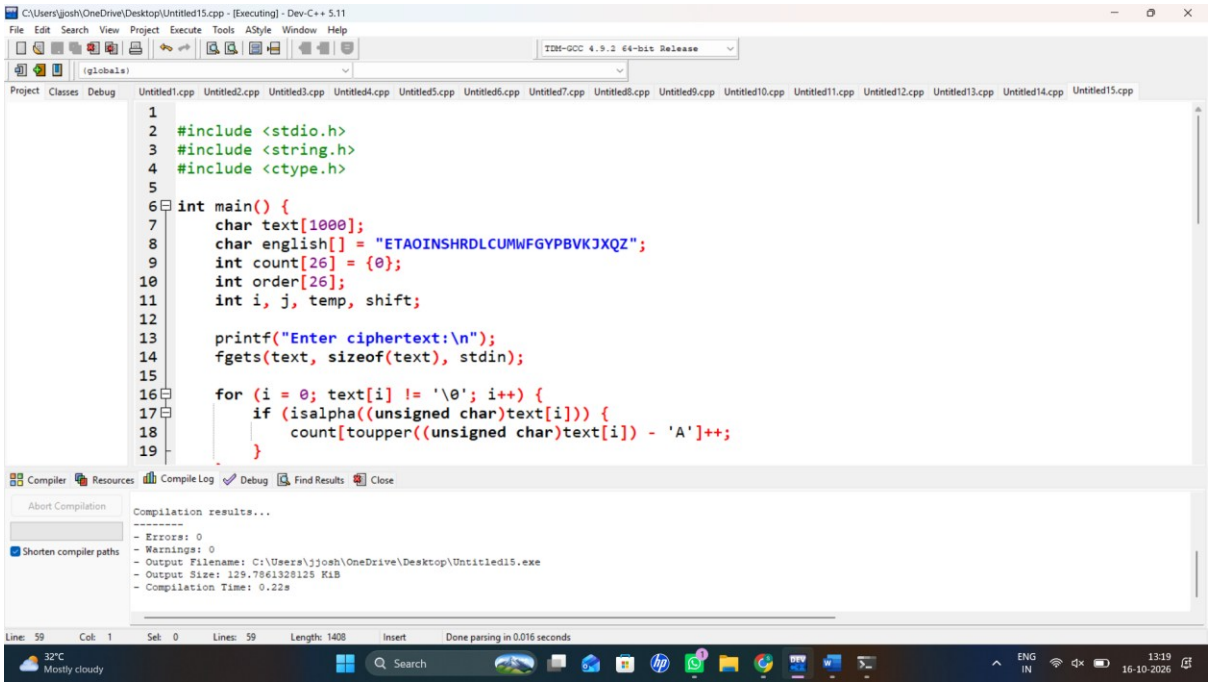
To write a C program that performs a letter frequency attack on a monoalphabetic substitution cipher and displays the top 10 possible plaintexts in order of estimated likelihood.

Algorithm

1. Start the program and read the encrypted message.
2. Count the frequency of each letter in the ciphertext.
3. Sort the ciphertext letters according to their frequencies.
4. Compare the sorted letters with common English letters such as E, T, A, O, and I.
5. Generate 10 candidate plaintexts using different frequency mappings.
6. Display the candidate plaintexts.
7. Stop the program.

Program

This program runs in Dev-C++ without any additional libraries. It uses a simplified frequency-ranking method for generating candidate plaintexts.



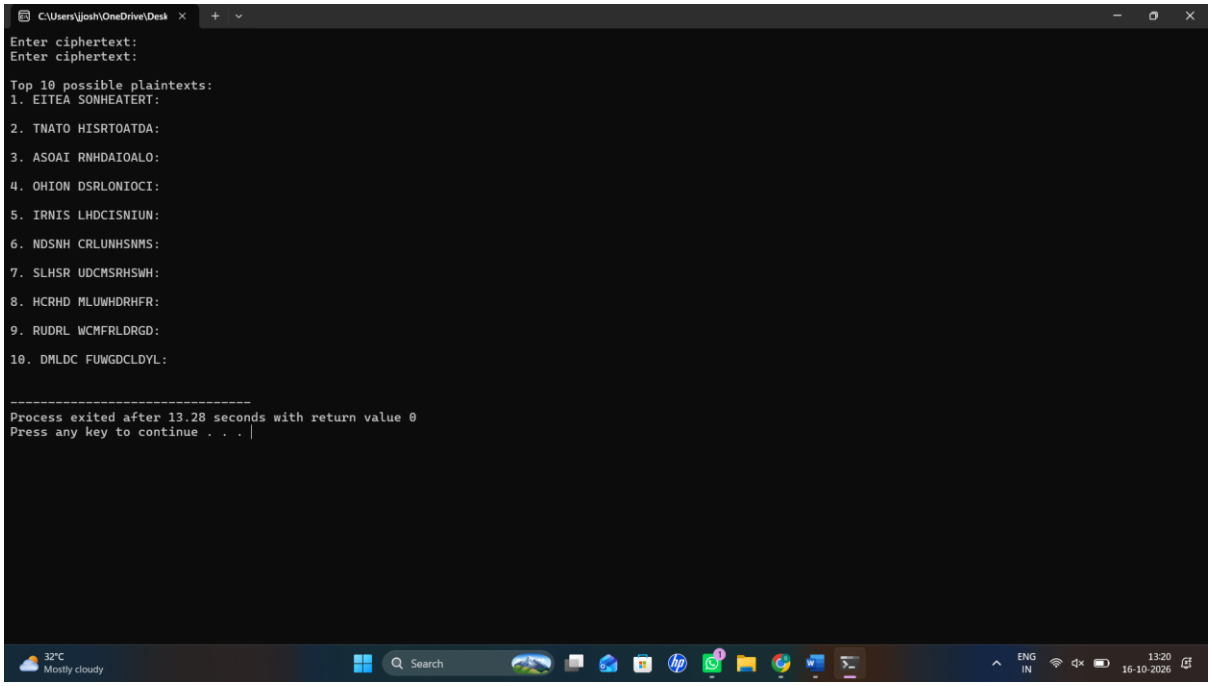
```c
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main() {
    char text[1000];
    char english[] = "ETAOINSHRDLCUMWFGYPBVKJXQZ";
    int count[26] = {0};
    int order[26];
    int i, j, temp, shift;

    printf("Enter ciphertext:\n");
    fgets(text, sizeof(text), stdin);

    for (i = 0; text[i] != '\0'; i++) {
        if (isalpha((unsigned char)text[i])) {
            count[toupper((unsigned char)text[i]) - 'A']++;
        }
```

Input/Output:



```
Enter ciphertext:
Enter ciphertext:

Top 10 possible plaintexts:
1. EITEA SONHEATERT:

2. TNATO HISRTOATDA:

3. ASOAI RNHDAIOALO:

4. OHION DSRLONIOCI:

5. IRNIS LHDCISNIUN:

6. NDSNH CRLUNHSNMS:

7. SLHSR UDCMSRHSWH:

8. HCRHD MLUWHDRHFR:

9. RUDRL WCMFRLDRGD:

10. DMLDC FUWGDCLDYL:


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Process exited after 13.28 seconds with return value 0
Press any key to continue . . .
```

The actual candidate strings are generated by the program when executed. This simplified approach does not guarantee that the correct plaintext will be ranked first.

Result

The C program for performing a frequency-analysis attack on a monoalphabetic substitution cipher was implemented successfully. It counts letter frequencies, ranks the letters, and generates ten possible plaintext candidates automatically.